**Society for Computer Technology & Research's (SCTR's)**

# Pune Institute of Computer Technology (PICT), Pune

**An Autonomous Institute affiliated to the Savitribai Phule Pune University (SPPU)**
**Approved by AICTE & Government of Maharashtra,**
**Accredited by NAAC (A+) & NBA [All eligible UG Programs]**

# Syllabus

# S.Y B. Tech Electronics and Telecommunication Engineering (E&TCE)

**With effect from (AY 2025-26)**
**National Education Policy (NEP) 2020 Compliant**
Approved by the Board of Studies (BoS E&TCE) and Academic Council

## Abbreviations used

| Sr. No. | Broad Category of the course | Sub- Category of course | Category Code |
|---|---|---|---|
| I. | Basic Science/ Engineering Science Course (BSC/ ESC) | Basic Science Course (BSC) | **01** |
| | | Engineering Science Course (ESC) | **02** |
| II. | Program Courses (PC) | Program Core Course (PCC) | **03** |
| | | Program Elective Course (PEC) | **04** |
| III. | Multidisciplinary Courses (MC) | Multidisciplinary Minor (MDM) | **05** |
| | | Open Elective (OE) | **06** |
| IV. | Skill Courses (SC) | Vocational and Skill Enhancement Course (VSEC) | **07** |
| V. | Humanities Social Science and Management (HSSM) | Ability Enhancement Course (AEC-01, AEC-02) | **08** |
| | | Entrepreneurship/Economics/ Management Courses (EEM) | **09** |
| | | Indian Knowledge System (IKS) | **10** |
| | | Value Education Course (VEC) | **11** |
| VI. | Experiential Learning Courses (ELC) | Research Methodology (RM) | 12 |
| | | Community Engagement Project (CEP) / Field Project (FP) | **13** |
| | | Project (PRJ) | **14** |
| | | Internship/ On Job Training (IP/OJT) | **15** |
| VII. | Liberal Learning Courses (LLC) | Co-curricular Activities (CCA) | **16** |
| VIII. | Honors Courses (HC) | offered within the same branch depth in same branch (advanced specialization & research). | **17** |
| IX. | Minor Courses (MIC) | offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).** | **18** |
| X. | Double Minor Courses (DMC) | Additional minor to be offered by other branches **breadth across disciplines (interdisciplinary exposure, applied projects).** | **19** |

**Detailed Guidelines for Examination:**

**Link:** Guidelines for examination

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# S.Y B. Tech Curriculum Structure

## Semester – III

| Semester -3 | | | Teaching Scheme (Hours/Week) | | | | Credit scheme | | | | Examination/ Evaluation Scheme and Marks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Theory | | | Practical | | | Total |
| | | | | | | | | | | | ISE | CIE | ESE | CIE | ESE | | |
| Category of Course | Course code | Name of the Course | L | P | T | Total | L | P | T | Total | [20] | [20] | [60] | TW | P | OR | |
| **PCC** | **2303101** | Signals and Systems (S&S) | 3 | - | 1 | **4** | 3 | - | 1 | **4** | 20 | 20 | 60 | 25 | - | 25 | **150** |
| **PCC** | **2303102** | Analog Circuit Design (ACD) | 3 | - | - | **3** | 3 | - | - | **3** | 20 | 20 | 60 | - | - | - | **100** |
| **PCC** | **2303203** | Analog Circuit Design Lab (ACDL) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | 25 | - | **50** |
| **PCC** | **2303104** | Network Analysis and Synthesis (NAS) | 2 | - | 1 | **3** | 2 | - | 1 | **3** | 20 | 20 | 60 | 25 | - | - | **125** |
| **VSEC** | **2307101** | Electronics Skill Development (ESD) | - | 4 | - | **4** | - | 2 | - | **22** | - | - | - | 50 | - | 25 | 75 |
| **MDM** | **03051X1** | MDM-1 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | 100 |
| **MDM** | **03052X1** | MDM-1 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | - | 25 | - | 25 |
| **OE** | **0306301** | OE-I: Foreign Language Studies (FLS) | - | - | 2 | **2** | - | - | 2 | **2** | - | - | - | 50 | - | - | 50 |
| **VEC** | **0311101** | Universal Human Values (UHV) | 2 | - | - | **2** | 2 | - | - | **2** | - | - | - | 25 | - | - | **25** |
| **AEC** | **0308202** | Professional Development and Career Readiness (PDCR) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| **CEP** | **03132XX** | Community Engagement project (CEP) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| | | **Total** | **12** | **12** | **4** | **27** | **12** | **6** | **4** | **22** | **80** | **80** | **240** | **250** | **50** | **50** | **750** |

**L: Lecture, P: Practical, T: Tutorial,**
**CIE: Continuous Internal Evaluation, ISE: In-Semester Examination, ESE: End-Semester Examination,**
**TW: Term work, OR: Oral, P: Practical**

# Semester – IV

| Semester-4 | | | Teaching Scheme (Hours/Week) | | | | Credit scheme | | | | Examination/ Evaluation Scheme and Marks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Category of Course | Course code | Name of the Course | L | P | T | Total | L | P | T | Total | Theory | | | Practical | | | Total |
| | | | | | | | | | | | ISE | CIE | ESE | CIE | ESE | | |
| | | | | | | | | | | | [20] | [20] | [60] | TW | P | OR | |
| PCC | 2403105 | Principles of Communication Engineering (PCE) | 3 | - | - | **3** | 3 | - | - | **3** | 20 | 20 | 60 | - | - | - | **100** |
| PCC | 2403206 | Principles of Communication Engineering Lab (PCEL) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | - | 25 | - | **25** |
| PCC | 2403107 | Digital Circuit Design (DCD) | 3 | - | - | **3** | 3 | - | - | **3** | 20 | 20 | 60 | - | - | - | **100** |
| PCC | 2403208 | Digital Circuit Design Lab (DCDL) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | 25 | - | **50** |
| PCC | 2403109 | Control Systems (CS) | 2 | - | 1 | **3** | 2 | - | 1 | **3** | 20 | 20 | 60 | 25 | - | - | **125** |
| VSEC | 2407202 | Project Based Learning (PBL) | - | 4 | - | **4** | - | 2 | - | **2** | - | - | - | 25 | - | 25 | **50** |
| EEM | 2409101 | Project Management and Finance Essentials (PMFE) | 2 | - | - | **2** | 2 | - | - | **2** | - | - | 50 | 25 | - | - | **75** |
| MDM | 04051X2 | MDM-2 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | **100** |
| MDM | 04052X2 | MDM-2 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | - | 25 | - | **50** |
| OE | 04063XX | Open Elective-II (OE-II) * | - | - | 2 | 2 | - | - | 2 | 2 | - | - | 50 | - | - | - | 50 |
| AEC | 0408203 | Collaborative Skills, Digital Ethics, and Cyber Security (CDC) | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| VEC | 0411102 | Indian Constitution and Social Responsibility (ICSR) | 1 | - | - | **1** | 1 | - | - | **1** | - | - | - | 25 | - | - | **25** |
| | | **Total** | **13** | **12** | **3** | **28** | **11** | **7** | **3** | **22** | **80** | **80** | **340** | **150** | **50** | **50** | **750** |

**#:** Tutorial or laboratory as applicable. Choose one course from the MDM baskets. MDM: X is basket number, **Refer Annexure-I** for MDM details.
*: Open Elective (OE) offered by online platform such as SWAYAM/ NPTEL, **Refer Annexure-II** for details.
XX: Serial number of the courses under that particular category.

# Second Year B-Tech
# (S. Y. B. Tech)
# **Semester-3**

**Second Year B. Tech (S. Y B. Tech) AY (2025-26)**
**Electronics and Telecommunication Engineering (E&TCE)**

## [2303101]: Signals and Systems (S&S)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 3 | L: 3 Hrs./ Week | CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks |
| | 1 | Tut: 1 Hr./ Week | ESE (OR): 25 Marks<br>CIE (TW): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Fundamentals of linear algebra, complex numbers, calculus, linear equations, partial differential equations.

**Course Objectives: The objective of this course is to provide students with**

- A fundamental understanding of signals and systems concepts and essential groundwork for advanced courses such as signal processing, control systems, and communication.
- Analysis of systems using impulse and step responses, convolution integrals, and convolution sums.
- Frequency-domain analysis for periodic and non-periodic signals through Fourier series and Fourier transforms.
- A foundational understanding of LTI system analysis using the Laplace transform.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Identify and classify** basic signals based on their mathematical descriptions and **perform** operations on dependent and independent variables of deterministic signals. **Apply** the properties of systems to characterize and **classify** them using input-output relations and impulse responses.

**CO2: Analyze** linear time-invariant systems to **determine** the output signal for a given system impulse response and arbitrary input using convolution integral and sum.

**CO3: Analyze** and **resolve** the signals in the frequency domain using Fourier analysis. **Determine** and **sketch** the amplitude and phase spectrum.

**CO4:** Given a time domain signal, **determine** the Laplace transform and **draw** its ROC. Given an impulse response of the LTI system, **analyze** the stability and causality using the Laplace transform and its properties.

## COURSE CONTENTS

| Module-I | Introduction to Signals and Systems | 10 Hrs. |
|---|---|---|

Signals: Introduction, Elementary signals: Unit impulse, Unit step, Unit ramp, Exponential, Sinusoidal, Rectangular pulse, Signum, and Sinc function. Operations on signals: time shifting, time reversal, time scaling, amplitude scaling, signal addition, and subtraction. Classification of signals: deterministic and random, periodic and non-periodic, energy and power, causal and non-causal, and even and odd signals. Systems: Introduction, Classification of systems: static and dynamic, causal and non-causal, linear and non-linear, time-variant and time-invariant, stable and unstable, and invertible and non-invertible systems. Analysis of the input voltage (unit step) and current signal in an RL/ RC circuit.

| Module-II | Time domain representation of LTI System | 10 Hrs. |
|---|---|---|
| Definition of the impulse response, Convolution integral: Computation of convolution integral using the graphical method for unit step to unit step, unit step to exponential, unit step to rectangular, and rectangular to rectangular signals. Convolution sum and its computation, Properties of convolution. System interconnection, System properties in terms of impulse response, Step response in terms of impulse response. Computing the impulse response of an RL/ RC circuit and using convolution to find its output for a rectangular voltage input. | | |
| **Module-III** | **Fourier Analysis of Signals** | **10 Hrs.** |
| Exponential Fourier Series, Symmetries in exponential Fourier series, Properties of Fourier series, Gibb's phenomenon. Fourier Transform (FT) of aperiodic continuous time (CT) signals, Dirichlet conditions for the existence of Fourier transform, FT of standard CT signals, Properties, and their significance, Interplay between time and frequency domain using Sinc and Rectangular signals, FT for periodic signals. Frequency analysis of an RL/ RC circuit using Fourier methods. | | |
| **Module-IV** | **Laplace Transform** | **09 Hrs.** |
| Definition of Laplace Transform (LT), Region of Conversion, Laplace transform of standard signals, Properties of ROC, Properties of Laplace transform and their significance, Inverse Laplace transform based on partial fraction expansion, Stability considerations in S domain, Application of Laplace transforms to the LTI system analysis. Application of Laplace transform to model and analyze a series RL/ RC circuit with a step input voltage. | | |
| **Text Books:** | | |
| **T1.** | S. Haykin and B. Van Veen, *Signals and Systems*, 2nd ed. New Delhi, India: Wiley India, 2007. | |
| **T2.** | C. Phillips, *Signals, Systems and Transforms*, 3rd ed. New Delhi, India: Pearson Education, 2003. | |
| **Reference Books:** | | |
| **R1.** | M. J. Roberts, *Signals and Systems*. New Delhi, India: Tata McGraw-Hill, 2007. | |
| **R2.** | N. Kanni, *Signals and Systems*, 2nd ed. New Delhi, India: McGraw-Hill, 2013. | |
| **Relevant MOOCs Course (Course name and Weblink)** | | |
| NPTEL Course: Principles of Signals and Systems, by Prof. Aditya K. Jagannatham, IIT Kanpur, Link: https://nptel.ac.in/courses/108104100. | | |
| **Relevant Topics for Self-study:** | | |
| Gaussian function, Signal multiplication, Convolution of exponential-to-exponential signal, Continuous-Time Fourier Series, Discrete-Time Fourier Series, Continuous-Time Fourier Transform, Discrete-Time Fourier Transform, Discrete Fourier Transform, Laplace Transform, Z-Transform, and their interrelations. | | |

**List of Tutorials:**

| Sr. No. | Problem Statement | COs |
|---|---|---|
| 1. | A. Sketch and write mathematical expressions for the following elementary signals in continuous time and discrete time:<br>a) Unit Impulse b) Unit Step c) Unit Ramp d) Rectangular<br>e) Sinusoidal f) Exponential g) Signum h) Sinc i) Triangular<br>B. Classify and find the respective value for the above signals.<br>a) Periodic / non-periodic b) Energy / Power/ Neither | CO1 |
| 2. | Consider any two continuous and discrete time signals and perform operations like amplitude scaling, addition, multiplication, differentiation, integration (accumulator for discrete time), time scaling, time shifting, and folding. | CO1 |
| 3. | Express the mathematical expressions of continuous time systems in input-output relation form and determine whether each is memoryless, causal, linear, stable, time-invariant, and invertible. | CO1 |
| 4. | Perform convolution two continuous-time signals using convolution integral and discrete-time signals using convolution sum. Prove the properties of convolution. | CO2 |
| 5. | Express the mathematical expressions of discrete time systems in impulse response form and determine whether each is memoryless, causal, linear, stable, and time-invariant. | CO2 |
| 6. | Compute Fourier Series components for the signals and plot their magnitude and phase response. | CO3 |
| 7. | State and prove the various properties of the Continuous Time Fourier Transform. Find the Fourier Transform of signals and plot amplitude and phase spectrum. | CO3 |
| 8. | Analyze the frequency response of an RL/RC circuit using the Fourier Transform and interpret the results. | CO3 |
| 9. | Compute the Laplace Transform of standard signals and determine the Region of Convergence (ROC). State and prove the properties of Continuous Time Laplace Transform. Find the Laplace Transform of given signals. | CO4 |
| 10. | Model and analyze a series RL/RC circuit with a step input voltage using the Laplace Transform and determine the output response. | CO4 |

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)
## [2303102]: Analog Circuit Design (ACD)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 3 | L: 3 Hrs./ Week | CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basic electronic circuit components such as resistors, capacitors, and diodes.
- Fundamentals of semiconductor physics.
- Basic principles of transistor operation.
- Network analysis techniques.

**Course Objectives: The objective of this course is to provide students with**

- An understanding of the construction, characteristics, and operation of MOSFETs, along with their applications in amplifier and switching circuits.
- The ability to analyze the functionality of MOSFET-based circuits, feedback amplifiers, and oscillators used in analog electronics.
- The skills to design and evaluate operational amplifier-based analog circuits for real-world applications, including signal conditioning, waveform generation, and analog-to-digital conversion.

**Course Outcomes: After completing this course, students will be able to:**

**CO1: Explain** the working principles of MOSFETs, their non-ideal characteristics, and their role in amplifier and switching circuits.

**CO2: Apply** MOSFETs and operational amplifiers to design and simulate basic analog electronic circuits.

**CO3: Analyze** the performance of different types of feedback amplifiers and oscillators based on circuit parameters and design specifications.

**CO4: Evaluate** and **design** operational amplifier-based applications such as comparators, rectifiers, and waveform generators for industrial and academic purposes.

## COURSE CONTENTS

| Module-I | MOSFET Basic | 10 Hrs. |
|---|---|---|

Enhancement MOSFET: Construction, Characteristics, DC Load line, AC equivalent circuit, Parameters, Parasitic. Non ideal characteristics: Finite output resistance, Body effect, Sub-threshold conduction, breakdown effects, temperature effect, effect of W/L ratio, Common source amplifier & analysis, Source follower: circuit diagram, comparison with common source, Frequency response for amplifier.

| Module-II | MOSFET Applications | 10 Hrs. |
|---|---|---|

Introduction to MOSFET as a basic element in VLSI, MOSFET as switch, CMOS inverter, resistor & diode. Current sink & source, Current mirror. Four types of feedback amplifiers, Effects of feedback, Voltage series & current series feedback amplifiers and analysis, Barkhausen criterion, Wein bridge & phase shift oscillator.

| Module-III | OP-AMP Basics | 10 Hrs. |
|---|---|---|

| | | |
|---|---|---|
| Block diagram, Differential amplifier analysis for Dual input Balanced output mode - AC analysis (using r parameters) & DC analysis, Level shifter, Op amp parameters, Current mirror, Op-amp characteristics (AC & DC). Voltage series & voltage shunt feedback amplifiers, Effect on Ri, Ro, gain & bandwidth. | | |
| **Module-IV** | **OP-AMP Applications** | **09 Hrs.** |
| Inverting amplifier, non-inverting amplifier, Voltage follower, Summing amplifier, Differential amplifier, Practical integrator, Instrumentation amplifier, Precision rectifier, Comparator, Schmitt trigger, Wave form generator (Square & triangular wave generator), DAC, ADC. | | |
| **Text Books:** | | |
| **T1.** | D. Neamen, *Electronic Circuits: Analysis and Design*, 3rd ed. New York, NY, USA: McGraw-Hill, 2002. | |
| **T2.** | R. A. Gaikwad, *Op-Amps and Linear Integrated Circuits*. New Delhi, India: Pearson Education, 2005. | |
| **T3.** | K. R. Botkar, *Integrated Circuits*. Mumbai, India: Technical Publications, 2010. | |
| **Reference Books:** | | |
| **R1.** | J. Millman and C. Halkias, *Integrated Electronics: Analog and Digital Circuits and Systems*. New Delhi, India: McGraw-Hill, 1991. | |
| **R2.** | D. A. Bell, *Electronic Devices and Circuits*, 5th ed. Oxford, U.K.: Oxford University Press, 2009. | |
| **R3.** | A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 5th ed. New York, NY, USA: Oxford University Press, 1999. | |
| **Relevant MOOCs Course (Course name and Weblink)** | | |
| 1. NPTEL Course: Analog Electronic Circuits, IIT Kharagpur, by Prof. Pradip Mandal<br>Link: https://nptel.ac.in/courses/108105158<br>2. NPTEL Course: **Analog Circuits, IIT Bombay, by Prof.** Jayanta Mukherjee<br>Link: https://nptel.ac.in/courses/108101094<br>**Virtual Laboratory Links:**<br>1. Integrated Circuits:<br>Link: http://vlabs.iitb.ac.in/vlabs-dev/vlab_bootcamp/bootcamp/electronerds/index.html<br>2. Basic Electronics Virtual Lab<br>Link: http://vlabs.iitkgp.ernet.in/be/ | | |
| **Relevant Topics for Self-study:** | | |
| Study Various types of BJT, JFET, D-MOSFET with their construction, Working and Q-point calculations. | | |

**Second Year B. Tech (S. Y B. Tech) AY (2025-26)**

**Electronics and Telecommunication Engineering (E&TCE)**

# [2303203]: Analog Circuit Design Lab (ACDL)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks<br>ESE (P): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basic electronic components and circuit theory.
- Fundamentals of MOSFET and Op-Amp operations.
- Circuit simulation tools such as Micro-Wind and SPICE.
- Basics of analog and digital electronics.

**Course Objectives: The objective of this course is to provide students with**

- The ability to apply fundamental concepts of MOSFETs and operational amplifiers to design and implement basic analog circuits.
- The skills to analyze the performance characteristics of MOSFET-based and Op-Amp-based circuits through simulation and experimental testing.
- The expertise to design and evaluate practical analog circuits, including amplifiers, Schmitt triggers, and digital-to-analog converters (DACs), for real-world applications.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Construct** and **test** various analog circuits such as amplifiers, Schmitt triggers, and DACs using MOSFETs and operational amplifiers.

**CO2: Analyze** the frequency response, gain, and impedance characteristics of single-stage amplifiers and differential amplifiers through simulations and experimental verification.

**CO3: Evaluate** the performance of Op-Amps by measuring parameters like input bias current, slew rate, and CMRR, and compare them with datasheet specifications.

## COURSE CONTENTS

| Expt. No. | Problem Statement | Hrs. | CO |
|---|---|---|---|
| 1. | Design, build single stage CS amplifier & verify dc operating point. | 2 | CO1 |
| 2. | Simulate single stage CS amplifier, plot frequency response. Calculate $A_v$, $R_i$, $R_o$ & bandwidth. | 2 | CO2 |
| 3. | Simulate CMOS inverter and transaction switch using Micro wind | 2 | CO2 |
| 4. | Measure following Op- amp parameters & compare with specifications given in data sheet.<br>a) Input bias current<br>b) Input offset current<br>c) Input offset voltage<br>d) Slew rate<br>e) CMRR | 2 | CO3 |
| 5. | Simulation of differential amplifier up to second stage using MOSFET | 2 | CO2 |
| 6. | Design, build & test integrator using Op-amp for given frequency $f_a$. | 2 | CO1 |
| 7. | Design, build & test Schmitt trigger using Op-Amp (LF356, TI071) | 2 | CO1 |

| | | | |
|---|---|---|---|
| 8. | Design, build & test 2 or 3-bit R-2R ladder DAC. | 2 | CO1 |
| 9. | Design PCB for any suitable circuit | 2 | CO1 |

**Text Books:**

| | |
|---|---|
| **T1.** | D. Neamen, Electronic Circuits: Analysis and Design, 3rd ed. New York, NY, USA: McGraw-Hill, 2002. |
| **T2.** | R. A. Gaikwad, Op-Amps and Linear Integrated Circuits. New Delhi, India: Pearson Education, 2005. |
| **T3.** | K. R. Botkar, Integrated Circuits. Mumbai, India: Technical Publications, 2010. |

**Reference Books:**

| | |
|---|---|
| **R1.** | J. Millman and C. Halkias, Integrated Electronics: Analog and Digital Circuits and Systems. New Delhi, India: McGraw-Hill, 1991. |
| **R2.** | D. A. Bell, Electronic Devices and Circuits, 5th ed. Oxford, U.K.: Oxford University Press, 2009. |
| **R3.** | A. S. Sedra and K. C. Smith, Microelectronic Circuits, 5th ed. New York, NY, USA: Oxford University Press, 1999. |

**Relevant Topics for Self-study:**
Transistor as a switch in CE configuration, Application of Schmitt trigger (Temperature controller), PID controller.

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)
### [2303104]: Network Analysis and Synthesis (NAS)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 2 | L: 2 Hrs./ Week | CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks |
| | 1 | Tut: 1 Hr./ Week | CIE (TW): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

Basics of Circuit Analysis: Network Terminology, Voltage and current sources, Current Division, Voltage Division.

**Course Objectives: The objective of this course is to provide students with**

- A fundamental understanding of circuit analysis and network design.
- Knowledge of different types of circuit analysis, simplification techniques, and network theorems essential for analyzing electrical networks.
- Proficiency in transient analysis techniques for R-L and R-C circuits.
- An introduction to two-port networks and their applications.
- The skills to design and implement filters for various signal processing applications.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Select** and **apply** the techniques & tools such as nodal analysis, mesh analysis, Thevenin's, Norton's, Superpositions, Maximum Power transfer theorems, and transformations to **solve** the electrical circuits.

**CO2: Formulate** and **analyze** the driven and source free RL and RC circuits. **Solve** for RL & RC circuits to **find** the transient responses in mathematical form.

**CO3: Formulate** the network equations and find the parameters for given network.

**CO4: Design** & **analyze** the filters for the given specifications.

### COURSE CONTENTS

| Module-I | Circuit Analysis and Network Theorems | 08 Hrs. |
|---|---|---|

Kirchoff's Laws, DC circuit analysis using mesh and node. Network theorems: Thevenin's, Norton's, Maximum power transfer & Superposition. Circuit simplification techniques: source transformation & source shifting.

| Module-II | Transient Analysis of Circuits | 07 Hrs. |
|---|---|---|

Time constant, initial & final conditions for circuit elements, transient analysis of R-L and R-C circuits, analysis of first ordered equations.

| Module-III | Two Port Parameters | 06 Hrs. |
|---|---|---|

Relationship of two port variables, short circuit admittance parameters, open circuit impedance parameters, transmission parameters, hybrid parameters, reciprocity and symmetry condition.

| Module-IV | Filters | 05 Hrs. |
|---|---|---|

Introduction to filters, Filter classification, filter parameters. Introduction to fundamental T & $\pi$ filter sections, design of Butterworth low pass filters, transformation of low pass filters to high pass filter, band pass filter, band stop filter.

**Text Books:**

**T1.** M. E. Van Valkenburg, *Network Analysis*. 3rd edition, New Delhi, India: Prentice-Hall of India,

**T2.** R. R. Singh, *Network Analysis & Synthesis*. New Delhi, 3rd edition, India: McGraw-Hill Education.

**T3.** W. H. Hayt, J. E. Kemmerly, and S. M. Durbin, *Electrical Circuit Analysis,* 6th edition, New Delhi, India: Tata McGraw-Hill.

**Reference Books:**

**R1.** F. F. Kuo, *Network Analysis and Synthesis,* 2nd edition, New York, NY, USA: Wiley International.

**R2.** S. Chatterjee, *Electrical Circuit Analysis & Network Theory,* 1st edition, New Delhi, India: All India Council for Technical Education (AICTE).

**R3.** N. C. Jagan and C. Lakshminarayana, *Network Theory*. 4th edition, Hyderabad, India: BS Publications.

**Relevant MOOCs Course (Course name and Weblink)**

1. NPTEL Course: Basic Electrical Circuits, IIT Madras, by Prof. Gajendranath Chowdary
   Link: https://nptel.ac.in/courses/117/106/117106108/
2. SWAYAM Course: Network Analysis, by Prof. Tapas Kumar Bhattacharya, IIT Kharagpur
   Link: https://onlinecourses.nptel.ac.in/noc20_ee46/preview

**Relevant Topics for Self-study:**

Transient analysis for RLC circuits.

**List of tutorials**:

| Sr. No. | Problem Statement | CO |
|---|---|---|
| 1. | Determine the network parameters using KVL, KCL, node analysis, mesh analysis and circuit simplification techniques. | CO1 |
| 2. | Determine the network parameters using Network Theorems | CO1 |
| 3. | Formulate differential equation for R-L circuit and solve for current and voltages by determining initial conditions for driven and source free conditions and verify the same using simulation. | CO2 |
| 4. | Formulate differential equation for R-C circuit and solve for current and voltages by determining initial conditions for driven and source free conditions and verify the same using simulation. | CO2 |
| 5. | Determine the short circuit admittance parameters, open circuit impedance parameters for a given network. | CO3 |
| 6. | Determine the transmission parameters, hybrid parameters for a given network. | CO3 |
| 7. | For the given Butterworth low pass filter specifications, find the order of the filter and show the structure. | CO4 |
| 8. | Convert the above Butterworth low pass filter into high pass filter, band pass filter, band stop filter using appropriate transformations. | CO4 |
| 9. | Simulation of RL, RC circuits using MATLAB / SCILAB. | CO2 |

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)
### [2307201]: Electronics Skill Development Lab (ESDL)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 2 | P: 4 Hrs. / Week | CIE (TW): 50 Marks<br>ESE (OR): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basics of electronics devices, circuits sensors.

**Course Objectives: The objective of this course is to provide students with**

- A fundamental understanding of Arduino microcontroller architecture and its programming environment.
- Hands-on experience in interfacing sensors, actuators, and communication modules for embedded applications.
- The ability to design and implement basic embedded system prototypes using Arduino.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Demonstrate** the ability to interface various sensors, actuators, and communication modules with the Arduino microcontroller.

**CO2: Implement** digital and analog input/output operations, PWM control, and serial communication to develop embedded system applications.

**CO3: Design** basic embedded system prototypes using Arduino for automation, measurement, and wireless communication applications.

**COURSE CONTENTS**

| Expt. No. | Problem Statement | Hrs. | CO |
|---|---|---|---|
| 1 | Introduction to Arduino and LED Blinking | 4 | CO1 |
| 2 | Switch/ Button-Controlled circuit (Digital Input and Output) | 2 | CO1 |
| 3 | Analog and Digital Sensor Interfacing | 2 | CO1 |
| 4 | DC Motor Speed Control | 2 | CO2 |
| 5 | Servo Motor Control | 2 | CO2 |
| 6 | LCD/GLCD/OLED Display Interfacing using I2C | 2 | CO2 |
| 7 | Distance Measurement | 2 | CO2 |
| 8 | Wireless Communication using Bluetooth Module | 2 | CO3 |


**Text Books:**

**T1.** Banzi, M., and Shiloh, M. *Getting Started with Arduino* 3rd ed. O'Reilly Media, 2014.

**T2.** Monk, S. *Programming Arduino: Getting Started with Sketches*, 2nd ed. McGraw Hill, 2013.

**T3.** Norris, T. *Arduino for Beginners: Essential Skills Every Maker Needs*. Que Publishing, 2016.

**Reference Books:**

**R1.** Purdum, J. *Beginning C for Arduino: Learn C Programming for the Arduino,* 2nd ed. Apress, 2020.

**R2.** McRoberts, M. *Beginning Arduino,* 2nd ed. Apress, 2013.

**R3.** Margolis, M. *Arduino Cookbook*, 3rd ed. O'Reilly Media, 2020.

| | |
|---|---|
| **R4.** | Cicci, J. *Arduino Project Handbook: 25 Practical Projects to Get You Started*. No Starch Press, 2017. |
| **Relevant MOOCs Course (Course name and Weblink)** | |
| | |
| **Relevant Topics for Self-study:** | |
| | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br><br>[03051X1]: Multidisciplinary Minor (MDM-1) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **3** | **2** | **L: 2 Hrs./ Week** | **CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks** |
| Refer Annexure-I | | | |
| Refer: Course list | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br><br>[03052X1]: Multidisciplinary Minor Lab/Tutorial (MDM-1) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **3** | **1** | **P/Tut: 2 Hrs./ Week** | **ESE (P): 25 Marks** |
| Refer Annexure-I | | | |
| Refer: Course list | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br><br>[03063XX]: OE-I Foreign Language Studies (FLS) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **3** | **2** | **Tut.: 2Hrs./ Week** | **CIE (TW): 50 Marks** |
| Refer Annexure-II for guideline | | | |
| Refer: Common Courses | | | |

**Second Year**
**B. Tech (S. Y B. Tech) AY (2025-26)**
**Electronics and Telecommunication Engineering (E&TCE)**

**[0311101]: Universal Human Values (UHV)**

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 2 | L: 2 Hrs. / Week | CIE (TW): 25 Marks |
| Refer: Common Courses | | | |

**Second Year B. Tech (S. Y B. Tech) AY (2025-26)**
**Electronics and Telecommunication Engineering (E&TCE)**

**[0308202]: Professional Development and Career Readiness (PDCR)**

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks |
| Refer: Common Courses | | | |

**Second Year B. Tech (S. Y B. Tech) AY (2025-26)**
**Electronics and Telecommunication Engineering (E&TCE)**

**[0313201]: Community Engagement Project (CEP)**

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 3 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks |
| Refer: Common Courses | | | |
| | | | |

# Second Year B. Tech
# (S.Y. B. Tech)
# **Semester-4**

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2403105]: Principles of Communication Engineering (PCE)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 3 | L: 3 Hrs./ Week | ISE: 20 Marks<br>CIE: 20 Marks<br>ESE: 60 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basics of Fourier analysis, Signals and Systems

**Course Objectives: The objective of this course is to provide students with**

- A fundamental understanding of communication systems essential for analyzing modern analog and digital communication technologies.
- Knowledge of various analog modulation schemes (AM, FM), the need for sampling and quantization in PCM, and different waveform coding techniques such as DM and ADM.
- An understanding of various line coding schemes and their appropriate applications.
- Insights into different digital modulation schemes and spread spectrum techniques.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Define** AM and FM techniques, analyze them in time and frequency domains, and explain their generation and detection methods. Compare power requirements, bandwidth, and hardware complexity.

**CO2: Explain** the sampling process and theorem for low-pass signals. Sketch the frequency spectrum for ideal, natural, and flat-top sampling. Draw and describe PCM, DM, and ADM modulators and demodulators.

**CO3: Compare** Polar, Unipolar, and Manchester line codes based on PSD, transparency, and error detection. Draw the transmitter and receiver block diagrams for BASK, BPSK, BFSK, QPSK, and MPSK, highlighting each block's function.

**CO4: Draw** the transmitter and receiver block diagrams for QASK, MSK, OFDM, DSSS, and FHSS. Analyze and compare bandpass modulation techniques based on BER, hardware complexity, and applications.

## COURSE CONTENTS

| Module-I | Analog Transmission & Reception | 10 Hrs. |
|---|---|---|

Amplitude modulation (DSB-FC), Double sideband Suppressed carrier (DSB-SC) modulation Spectrum and Bandwidth of AM, DSB-SC, Calculation of Modulation Index for AM wave, Power and power efficiency, Block diagram of AM receiver. Frequency Modulation (FM), Modulation Index, Spectrum of FM (single tone): Feature of Bessel Coefficient, Power of FM signal, Bandwidth of FM signal, FM Modulator, FM generation by Armstrong's Indirect method, FM demodulator.

| Module-II | Pulse Modulation | 10 Hrs. |
|---|---|---|

Sampling theorem for low pass signal in time domain and Fourier domain and Nyquist criteria, Types of sampling- natural and flat top. Pulse amplitude modulation & concept of TDM: Channel bandwidth for PAM, Quantization of Signals, Quantization error, Companding: A-law & μ-law. Generation &

| | | |
|---|---|---|
| Reconstruction of Pulse code modulation (PCM), Differential Pulse code modulation, Delta Modulation, Adaptive Delta Modulation. | | |
| **Module-III** | **Digital Modulation I** | **10 Hrs.** |
| Line codes: Properties and spectrum, Baseband Signal Receiver, Digital Modulation: Generation, Reception, Signal Space Representation and Probability of Error Calculation for Binary Phase Shift Keying (BPSK), Binary Frequency Shift Keying (BFSK), Quadrature Phase Shift Keying (QPSK), M-ary Phase Shift Keying (MPSK). | | |
| **Module-IV** | **Digital Modulation II** | **09 Hrs.** |
| Generation, Reception, Signal Space Representation and Probability of Error Calculation for Quadrature Amplitude Shift Keying (QASK), Minimum Shift Keying (MSK), Orthogonal Frequency Division Multiplexing (OFDM), Comparison of digital modulation systems. Basics od Spread spectrum, Block diagram of Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread spectrum (FHSS). | | |
| **Text Books:** | | |
| **T1.** | B.P. Lathi, Zhi Ding, *Modern Analog and Digital Communication Systems*, 4th Edition, Oxford University Press, **2010**. | |
| **T2.** | Taub, Schilling, Saha, *Principles of Communication Systems*, 4th Edition, McGraw-Hill Education, **2013**. | |
| **Reference Books:** | | |
| **R1.** | Bernard Sklar, Prabitra Kumar Ray, Digital Communications: Fundamentals and Applications, 2nd Edition, Pearson Education, 2009. | |
| **R2.** | Simon Haykin, Communication Systems, 4th Edition, John Wiley & Sons, 2001. | |
| **R3.** | A.B. Carlson, P.B. Crilly, J.C. Rutledge, Communication Systems, 5th Edition, Tata McGraw-Hill, 2010. | |
| **Relevant MOOCs Course (Course name and Weblink)** | | |
| 1. NPTEL Course: Principles of Communication Systems-I, by Prof. Aditya K. Jagannatham, IIT Kanpur, Link: https://nptel.ac.in/courses/108/104/108104091 <br>2. NPTEL Course: Principles of Communication, by Prof. V. Venkat Rao, IIT Madras. Link: https://nptel.ac.in/courses/117/106/117106090/ | | |
| **Relevant Topics for Self-study:** | | |
| AM receivers, Optimum Receiver, Mary-FSK | | |

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2403206]: Principles of Communication Engineering Lab (PCEL)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks<br>ESE(P): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basics of Fourier analysis, Signals and Systems

**Course Objectives: The objective of this course is to provide students with**

- A foundational understanding of analog modulation techniques such as Amplitude Modulation (AM) and Frequency Modulation (FM), including their generation, transmission, and reception through practical implementations.
- The knowledge and skills to analyze and implement digital pulse modulation techniques such as PCM and PAM, and various line coding schemes, along with the ability to evaluate their signal characteristics and bandwidth requirements.
- Practical exposure to digital modulation techniques including BFSK and DSSS, and to assess their performance under real-world conditions, including noise and interference, using simulation tools or hardware setups.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Analyze** and **demonstrate** the generation and reception of analog modulation schemes such as AM and FM, and evaluate their characteristics using waveform and spectrum observations.

**CO2: Construct**, **illustrate**, and **evaluate** digital pulse modulation and line coding schemes including PCM, PAM, and various line codes; analyze their bandwidth and signal characteristics.

**CO3: Demonstrate** and **assess** the performance of baseband and bandpass digital modulation techniques like BFSK and DSSS in the presence of noise using simulation or hardware tools.

### COURSE CONTENTS

| Expt. No. | Problem Statement | Hrs. | CO |
|---|---|---|---|
| 1. | Analyze AM transmitter and receiver using block diagrams; generate DSB-FC AM signals, compute modulation index and power, and evaluate waveform and spectral changes across various modulation indices. | 4 | CO1 |
| 2. | Analyze frequency modulator and demodulator systems using block diagrams; generate FM signals, compute modulation index and bandwidth, and interpret waveform and spectral characteristics. | 2 | CO1 |
| 3. | Apply the Sampling Theorem to generate PAM (natural and flat-top) signals, reconstruct the original waveform, and evaluate aliasing effects in the frequency domain. | 2 | CO2 |
| 4. | Construct and analyze a PCM system, sketch PCM waveforms, and compute signaling rate and bandwidth. | 2 | CO2 |

| | | | |
|---|---|---|---|
| **5.** | Generate and compare line codes (NRZ, RZ, AMI, Manchester), sketch corresponding waveforms, and analyze bandwidth requirements. | 2 | CO2 |
| **6** | Generate baseband input bit sequences and evaluate receiver performance in noisy environments using appropriate hardware setup. | 2 | CO3 |
| **7.** | Analyze BFSK modulation using block diagrams; generate ASK1, ASK2, BFSK waveforms, sketch input and carrier signals, and evaluate bandwidth using practical methods. | 2 | CO3 |
| **8.** | Demonstrate DSSS modulation; generate PN codes and DSSS signals using hardware setup and sketch the associated waveforms. | 2 | CO3 |

**Text Books:**

**T1.** B.P. Lathi, Zhi Ding, Modern Analog and Digital Communication Systems, 4$^{th}$ Edition, Oxford University Press, 2010.

**T2.** Taub, Schilling, Saha, Principles of Communication Systems, 4$^{th}$ Edition, McGraw-Hill Education, 2013.

**Reference Books:**

**R1.** Bernard Sklar, Prabitra Kumar Ray, Digital Communications: Fundamentals and Applications, 2$^{nd}$ Edition, Pearson Education, 2009.

**R2.** Simon Haykin, Communication Systems, 4th Edition, John Wiley & Sons, 2001.

**R3.** A.B. Carlson, P.B. Crilly, J.C. Rutledge, Communication Systems, 5$^{th}$ Edition, Tata McGraw-Hill, 2010.

**Relevant Topics for Self-study:**

Simulation of OFDM can be explored using Octave/ LabView

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2403107]: Digital Circuit Design (DCD)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 3 | L: 3 Hrs./ Week | CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks |

**Prerequisite: Students should have prior knowledge of**

- Knowledge about basic logic gates and its truth table, Implementation of simple combinational digital circuit using logic gate, Knowledge of different Number system and its conversion.
- Knowledge about basic Boolean laws and De morgans law for designing combinational and sequential circuits.

**Course Objectives: The objective of this course is to provide students with**

- A foundational understanding of two-valued logic and logical circuits.
- Knowledge of Boolean algebra, Karnaugh maps, and their applications in designing and analyzing digital circuits.
- The ability to design, implement, and verify logical operations using combinational and sequential logic circuits.
- An introduction to the basics of Hardware Description Language (HDL) and programmable logic devices.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Explain** and **compare** the characteristics of the logic families and **draw** the interfacing circuits (TTL to CMOS and vice-versa).

**CO2: Design, Analyze** and **Implement** the various combinational and sequential logic circuits to verify their function table.

**CO3: Define** and **compare** the state machine and **develop** the FSM using Mealy Moore machine representations.

**CO4: Design** of combinational circuits using various types of PLDs. **Design** the combinational logic circuits using HDL and PLDS.

### COURSE CONTENTS

| Module-I | Combinational Logic Design | 10 Hrs. |
|---|---|---|

Standard representation of logic functions, Realization of SOP and POS forms, Canonical form, Minimization of logic functions using K-map up to 4 variables. Design of Adders, Sub tractors, Code converters, Digital Comparators, Multiplexers, Demultiplexers, Decoders, Parity generator, Arithmetic Logic Unit

| Module-II | Sequential Logic Design | 10 Hrs. |
|---|---|---|

Flip Flops, Clocked SR, JK, T, D and MS-JK flip-flop, Excitation table for flip-flops, Conversion of flip-flops, Applications of flip-flops: Counters: Synchronous and Asynchronous counters, shift registers, sequence generators. Mealy and Moore machines representation.

| Module-III | ASM Design and Introduction to HDL | 10 Hrs. |
|---|---|---|

| | | |
|---|---|---|
| State diagram, State table, Design of State Machines using State assignment and State reduction, Design of sequence detector using Finite State Machine (FSM), Applications of FSM<br>**Introduction to HDL**, Modelling Styles, Modelling Combinational Logic using HDL, Modelling Sequential Logic using HDL. | | |
| **Module-IV** | **Digital Logic Families and PLD** | **09 Hrs.** |
| Classification of logic families, Characteristics of digital logic families: Speed of operation, Power Dissipation, Figure of merit, Fan in, Fan out, Current and Voltage parameters, Noise immunity, Operating temperatures and Power supply, Two Input TTL NAND Gate, CMOS Invertor, NAND and NOR, Interfacing<br>**Introduction to PLDs** and their types: ROM, PAL, PAL, CPLD and FPGA. | | |
| **Text Books:** | | |
| **T1.** | R.P. Jain, Modern Digital Electronics, 4th Edition, 12th Reprint, TMH Publication, 2007. | |
| **T2.** | Thomas Floyd, Digital Electronics, 11th Edition, Pearson Publication, Year not specified, 2017. | |
| **T3.** | M. Morris Mano, Digital Logic and Computer Design, 4th Edition, Prentice Hall of India, 2018. | |
| **T4.** | Taub and Schilling, Digital Principles and Applications, 7th Edition, Tata McGraw-Hill Education, 2010. | |
| **T5.** | S. Palnitkar, Verilog HDL – A Guide to Digital Design and Synthesis, 3rd Edition, Pearson Publication, 2010. | |
| **Reference Books:** | | |
| **R1.** | A. Anand Kumar, "Fundamentals of digital circuits" 4th edition, PHI publication, 2014. | |
| **R2.** | John F. Wakerly, "Digital Design: Principles and Practices", 3rd edition, 4th reprint, Pearson Education, 2004. | |
| **R3.** | M. M. Mano, "Digital Design," 6th Edition. Pearson Education, 2018. | |
| | | |
| **Relevant MOOCs Course (Course name and Weblink)** | | |
| 1. NPTEL Course: Digital Circuits and Systems, by Prof. Shankar Balchandran, IIT, Bombay.<br>Link: https://nptel.ac.in/courses /117/106/117106086/<br>2. NPTEL Course: Hardware modeling using verilog, by Prof. Indranil Sengupta, IIT Kharagpur,<br>Link: https://nptel.ac.in/courses/106/105/106105165/. | | |
| **Relevant Topics for Self-study:** | | |
| Quine-McCluskey Method 5-6 variables K-map, Ring / Twisted Ring counter using Shift Register. | | |

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2403208]: Digital Circuit Design Lab (DCDL)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks<br>ESE (P): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basic Boolean laws, De Morgan's law.
- Implementation of simple combinational digital circuit using logic gate.

**Course Objectives: The objective of this course is to provide students with**

- The ability to design, implement, and verify logical operations using combinational and sequential logic circuits.
- Skills to analyze digital circuits for functionality and performance, as well as to troubleshoot and resolve issues in digital designs.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Design & implement** various combinational and sequential logic circuits on digital trainer kit to verify their function table.

**CO2: Design & implement** various combinational and sequential logic circuits using HDL.

### COURSE CONTENTS

| Expt. No. | Problem Statement | Hrs. | CO |
|---|---|---|---|
| **1.** | **i). Study of IC-74LS153 as a Multiplexer.**<br>a) Design and Implement 8:1 MUX using IC-74LS153 & Verify its Truth Table.<br>b) Design & implement the given 4 variable functions using IC74LS153. Verify its Truth- Table.<br>**ii). Study of IC-74LS138 as a Demultiplexer / Decoder**.<br>a) Design and implement full adder and subtractor function using IC- 74LS138.<br>b) Design & Implement 3-bit code converter using IC-74LS138. (Gray to Binary/Binary to Gray) | 4 | CO1 |
| **2.** | **Study of IC-74LS83 as a BCD adder: (Refer Data-Sheet).**<br>a) Design and Implement 1-digit BCD adder usingIC-74LS83.<br>b) Design and Implement 4-bit Binary sub tractor using IC-74LS83. | 2 | CO1 |
| **3.** | **Study of IC-74LS85 as a magnitude comparator, (Refer Data-Sheet).**<br>Design and Implement 4-bit / 8-bit Comparator circuits. | 2 | CO1 |
| **4.** | **Study of Counters:**<br>Design and Implement 4-bit counter using JK- Flip flop. | 2 | CO1 |
| **5.** | **Study of Counter ICs (74LS90/74HC191): (Refer Data-Sheet)** | 4 | CO1 |

| | | | |
|---|---|---|---|
| | a) Design and Implement MOD-N and MOD-NN using IC-74LS90 and IC-74LS93.<br>b) Design & Implement MOD-N Up/down Counter using IC74HC191 / IC74HC193. | | |
| **6.** | Study of shift register: (IC75HC 194/IC74LS194)<br>a) Design and implement 4-bit register using D FF<br>b) Ring counter and twisted ring counter | 2 | CO1 |
| **7.** | Design and Implement Combinational Logic Circuit Using HDL. | 2 | CO2 |
| **8.** | Design and Implement Sequential Logic Circuit Using HDL. | 2 | CO2 |

**Text Books:**

**T1.** R.P. Jain, Modern Digital Electronics, $4^{th}$ Edition, $12^{th}$ Reprint, TMH Publication, 2007.

**T2.** Stephen Brown, *Fundamentals of Digital Logic Design with VHDL*, $1^{st}$ Edition, TMH Publication, **2002**.

**Reference Books:**

**R1.** A. Anand Kumar, "Fundamentals of digital circuits" $4^{th}$ edition, PHI publication, 2014.

**R2.** M. M. Mano, "Digital Design," $6^{th}$ Edition. Pearson Education, 2018.

**Relevant Topics for Self-study:**

Verify & compare four voltage and current parameters for TTL and CMOS (IC 74LSXX, 74HCXX), (Refer Data-Sheet).

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2403109]: Control Systems (CS)

<table>
<tr><th>Semester</th><th>Credits</th><th>Teaching Scheme</th><th>Examination Scheme</th></tr>
<tr><td rowspan="2">4</td><td>2</td><td>L: 2 Hrs./ Week</td><td>CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks</td></tr>
<tr><td>1</td><td>Tut: 2 Hrs./ Week</td><td>CIE(TW): 25 Marks</td></tr>
</table>

**Prerequisite: Students should have prior knowledge of**

- Concept of open loop and closed loop transfer function,
- Steady state and transient state, Pole- zero plot of a system.

**Course Objectives: The objective of this course is to provide students with**

- A comprehensive understanding of control system fundamentals, including various modeling techniques and analysis methods.
- The ability to evaluate system stability and performance through time and frequency response analysis.
- Knowledge of advanced control concepts such as root locus, Bode plots, state variable analysis, and PID controllers.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Classify** the control systems. **Obtain** the transfer function of a control system using block diagram reduction technique. **Define** stability and comment on the stability of the given system using Routh-Hurwitz Criterion.

**CO2: Analyze** the time domain response of a control system to obtain performance parameters like steady state error, static error coefficients, rise time, peak time, peak overshoot, settling time & delay time.

**CO3: Sketch** and **analyze** the root locus and comment on closed loop stability of a control system. **Determine** frequency domain parameters such as resonant frequency, resonant peak and bandwidth. **Draw** Bode plot and **analyze** the closed loop stability of a system by calculating gain margin and phase margin.

**CO4: Obtain** state equations, state diagram, state transition matrix and canonical forms for a control system using state space method. **Analyze** the characteristics of PID controller in P, I, D, PI, PD & PID modes. **State** the applications of PID controller.

## COURSE CONTENTS

| Module-I | Title: Introduction, Transfer Function and Stability of Control Systems | 08 Hrs. |
|---|---|---|

Basic Elements of Control System, Open loop and Closed loop systems, Differential equations and Transfer function, Block diagram reduction Techniques.

Characteristic equation of a system, concept of pole and zero, response of various pole locations in s-plane, concept of stability absolute stability, relative stability of system from pole locations, Routh Hurwitz stability criterion.

| Module-II | Title- Time Domain Analysis | 05 Hrs. |
|---|---|---|
| Time domain analysis: transient response and steady state response, standard test inputs for time domain analysis, order and type of a system, transient analysis of first and second order systems, time domain specifications of second order under damped system from its step response, Steady state error and static error constants. | | |
| **Module-III** | **Title- Frequency Response and Stability Analysis Techniques** | **08 Hrs.** |
| Root locus: definition, magnitude and angle conditions, construction of root locus, concept of dominant poles, effect of addition of pole and zero on root locus. Application of root locus for stability analysis. Frequency response and frequency domain specifications, stability analysis using Bode plot, Calculation of gain margin and phase margin from Bode plot. | | |
| **Module-IV** | **Title- State Space Analysis and Controllers** | **05 Hrs.** |
| State space analysis, advantages and representation, transfer function from State space, canonical forms, Solution of homogeneous state equations, state transition matrix and its properties, computation of state transition matrix by Laplace transform method only.<br>Concept of Controller, Basic ON-OFF Controller, Concept of Dead Zone, Introduction to P, I, D, PI, PD and PID controller, OFFSET of Controller, Integral Reset, PID Characteristics. | | |

**Text Books:**


**T1.** N. J. Nagrath & M. Gopal, *Control System Engineering*, 5th Edition, New Age International Publishers, 2017.

**T2.** K. Ogata, *Modern Control Engineering*, 5th Edition, Prentice Hall India Learning Private Limited, 2010.


**Reference Books:**


**R1.** Benjamin C. Kuo, Automatic Control Systems, 7th Edition, Prentice Hall of India, 2003.

**R2.** M. Gopal, Control System – Principles and Design, 4th Edition, Tata McGraw-Hill, 2012.

**R3.** Schaum's Outline Series, Feedback and Control Systems, Tata McGraw-Hill, 2010


**Relevant MOOCs Course (Course name and Weblink)**

1. NPTEL Course: Control systems, by Prof. C.S.Shankar Ram IIT Madras,
   Link: https://nptel.ac.in/courses/107/106/107106081/ .
2. NPTEL Course: Control System Design, by Prof. G R Jayanth, IISc Bangalore
   Link: https://nptel.ac.in/courses/115/108/115108104/

**Relevant Topics for Self-study**

Transfer function of RLC network, Mathematical Modeling of mechanical systems

**List of Tutorials:**

| Expt. No. | Problem Statement | CO |
|---|---|---|
| **1.** | Derive the transfer function of electrical and mechanical systems by applying mathematical modeling techniques using Force-Voltage and Force-Current analogies. | CO1 |
| **2.** | Evaluate the steady state error and calculate static error coefficients for the given control system. | CO2 |
| **3.** | Determine the time domain performance specifications including rise time, peak time, peak overshoot, settling time, and delay time for a given system. | CO2 |
| **4.** | Calculate and interpret the frequency domain specifications such as resonant frequency, resonant peak, and bandwidth for the given control system. | CO3 |
| **5.** | Sketch the Bode plot for a given open-loop transfer function G(s)H(s), and determine gain margin, phase margin, and assess the stability of the system. | CO3 |
| **6.** | Formulate the state equations, draw the state diagram, and compute the state transition matrix for a control system using the state-space representation. | CO4 |
| **7.** | Simulate a control system using software tools and analyze its performance parameters based on time and frequency domain responses. | CO2, CO3 |
| **8.** | Develop and execute a simulation program for a given control system using an object-oriented programming language to evaluate system behavior. | CO4 |

# Second Year B. Tech (S. Y B. Tech) AY (2025-26)
# Electronics and Telecommunication Engineering (E&TCE)

## [2407202]: Project Based Learning (PBL)

(Circuit Minds: Learn by Doing)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 1 | P: 2 Hrs./ Week | CIE (TW): 25 Marks<br>ESE (OR): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Basics of electronics components, circuits, electrical and electronics circuit analysis.
- C/ C++/ object-oriented programming and other programming knowledge.

**Course Objectives: The objective of this course is to provide students with**

- The ability to solve real-world problems individually or in groups using available resources.
- Skills to develop applications by applying electronics and communication engineering concepts, integrating prior knowledge when necessary.
- Hands-on experience in all stages of electrical and electronic system development, including specification, design, implementation, and testing.

**Course Outcomes: After studying this course students will be able to**

**CO1: Formulate** and **present** a project idea based on interest, literature survey, recent trends and real-life problems**. Plan** project work in team.

**CO2: Implement** electronic hardware by learning PCB artwork design, soldering techniques, testing, and troubleshooting etc. **Identify** appropriate solution and implement it using electronic hardware/software principles. **Demonstrate** the use of modern tools for simulation and implementation of the system.

**CO3: Prepare** a technical report based on the mini project work. **Comprehend** and **write** a project report and **draw** conclusions at a technical level.

## COURSE GUIDELINES

**A. Group Formation:**

- Form a group of 3-4 students that share a similar interest in each batch.
- The group should be cohesive, sharing and caring, contribute to the task assigned.
- The task allocation for each week should be maintained in LOG book by each group.
- Hardware projects should be encouraged (80%) and some software projects may be allowed (20%).

**B. Problem statement selections:**

- Each course teacher will provide a list of problems statement in particular course studying in current year. These statements will be displayed prior to the commencement of semester.
- Students are instructed to choose one out of the provided statements. The statement will be approved by course teachers.

**OR**

- A group of students will find THREE problem statements in any domain. Course teachers will approve one out of that depending on resources availability, and need of time. You may use following list to search appropriate project title.
  - Professional society (IEEE, IET, ACM etc.) Journal, Conference/transaction papers.
  - Electronics project or design magazine (E4U, ED, ESD etc.)
  - Component manufactures web sites (on semi, national semiconductors)
  - Data sheets/ application notes/ data manuals by electronics component manufacturers.
  - Design tutorials by electronics manufacturer.
  - Appendix, exercise section of reference books listed in the syllabus.
  - Recent trends in electronics.
  - Manufacturer challenges/ competition.
  - Carry out survey to solve problem by electronic means.
  - Robotics/ Robocon and other professional society requirements.
  - Extension to the old projects.
  - Social, live, sponsored, consultancy projects, inter-disciplinary may be encouraged.

**C. Evaluation Method:**

- The project Seminar-I (Introductory seminar) and Seminar-II (Completion seminar) are compulsory.
- Course teacher will prepare rubrics for the assessment and share the same with students at the commencement of semester.
- Week wise assessment is considered under the head continuous internal evaluation (CIE).

**D. Week wise Assessment schedule:**

| Week Scheduled | Task to be performed |
|---|---|
| Week-1 | a. Formation of Group and<br>b. Literature Survey, Finalizing the Specifications |
| Week-2 | a. Finalization of project titles<br>b. Seminar-I (Project Idea) Presentation |
| Week-3 | a. Selection of Components/devices/ algorithms, Paper Design<br>b. Block schematic and Circuit diagram/ flow charts |
| Week-4 | a. Simulation of Different modules/ functions<br>b. Component Purchasing, Breadboard testing/ PCB layout design.<br>c. Algorithm, Flow Chart testing |
| Week-5 | Programming, Assembling, Soldering and testing. |
| Week-6 | a. Integrating modules in HW/SW<br>b. Designing enclosures |
| Week-7 | a. Testing and Troubleshooting of HW/SW<br>b. Seminar –II (Project Work) Presentation |
| Week-8 | a. Testing and Troubleshooting of HW/SW<br>b. Seminar –II (Project Work) Presentation |
| Week-9 | a. Project Demonstration<br>b. Project report preparation |
| Week-10 | a. Project Exhibition<br>b. Final report submission |

***Note:*** Students are instructed to adhere to the schedule strictly to smooth conduction of course.

| **Reference Books:** | |
|---|---|
| **R1.** | Larmer, J., Mergendoller, J. R., & Boss, S., *Setting the Standard for Project Based Learning*, ASCD, 2015. |
| **R2.** | Larmer, J., & Boss, S., *Project Based Teaching: How to Create Rigorous and Engaging Learning Experiences*, ASCD, 2018. |
| **R3.** | Murphy, E. M., & Cooper, R., *Hacking Project Based Learning: 10 Easy Steps to PBL and Inquiry*, Times 10 Publications, 2017. |
| **R4.** | Krašna, M., *Project Based Learning (PBL) in the Teachers' Education*, 39th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), IEEE, pp. 852–856, 2016. |
| **R5.** | Macias-Guarasa, J., Montero, J. M., San-Segundo, R., Araujo, A., & Nieto-Taladriz, O., *A Project-Based Learning Approach to Design Electronic Systems Curricula*, IEEE Transactions on Education, Vol. 49, No. 3, pp. 389–397, 2006. |
| **Relevant MOOCs Course (Course name and Weblink)** | |
| SWYAM: Problem Based learning, by Dr. Indrajit Saha, National Institute of Technical Teachers Training and Research, Kolkata<br>Link: https://onlinecourses.swayam2.ac.in/ntr20_ed12/preview. | |

## Second Year B. Tech (S. Y B. Tech) AY (2025-26)
## Electronics and Telecommunication Engineering (E&TCE)

### [2409101]: Project Management and Finance Essentials (PMFE)

| Semester | Credits | Teaching Scheme | Examination Scheme |
|---|---|---|---|
| 4 | 2 | L: 2 Hrs./ Week | ESE: 50 Marks<br>CIE (TW): 25 Marks |

**Prerequisite: Students should have prior knowledge of**

- Principles of Management Course

**Course Objectives: The objective of this course is to provide students with**

- Fundamental concepts of project management and financial planning.
- An understanding of management evolution, principles, and strategic planning.
- Key aspects of forecasting, project estimation, and risk assessment.
- Enhanced decision-making abilities and organizational effectiveness.

**Course Outcomes: After completing this course, students will be able to**

**CO1: Describe** fundamental management principles and project management concepts, and **analyze** their applications in real-world scenarios.

**CO2: Explain** financial planning, risk assessment, and decision-making processes, and **evaluate** their effectiveness in project execution.

**CO3: Apply** basic management and project planning techniques to solve engineering and business-related problems, and **create** structured project plans for practical implementation.

### COURSE CONTENTS

| Module-I | Management and Project Fundamentals | 07 Hrs. |
|---|---|---|

- **Management Principles**: Definition, Nature, Scope, Characteristics, Functions, Roles, and Skills of an Effective Manager.
- **Evolution of Management**: Classical Theory, Scientific Management, Bureaucracy, Behavioral Science Approach, Systems Approach.
- **Project Management**: Introduction to Project Management, Project Life Cycle, Organization Strategy, and Project Selection.
- **Organizational Structure**: Project Management Organization Structure, Work Breakdown Structure (WBS).

| Module-II | Planning, Forecasting, and Risk Management | 06 Hrs. |
|---|---|---|

- **Planning**: Types of Plans, Planning Process, Strategic Management, Environmental Appraisal, Industry Analysis.
- **Forecasting**: Components of Business Forecasting, Benefits, Techniques, and Limitations.
- **Project Estimation**: Time & Cost Estimation, Network Analysis using PERT/CPM, Resource Levelling, Scheduling.
- **Project Risk Management**: Risk Identification, Quantification, Mitigation, and Capital Project Risk Assessment.

| Module-III | Financial and Objective Management | 07 Hrs. |
|---|---|---|

| | | |
|---|---|---|
| • **Decision-Making**: Decision-making Process, Group Decision-making, Problem-solving.<br>• **Management by Objectives (MBO)**: Concepts, Characteristics, Goal Setting, Action Plan.<br>• **Financial Management**: Profit Maximization, Wealth Maximization, Investment, Financing, and Dividend Decisions.<br>• **Investment Decisions**: Cost of Capital, Payback Period, Net Present Value, Internal Rate of Return, Profitability Index. | | |
| **Module-IV** | **Communication and Project Appraisal** | **06 Hrs.** |
| • **Communication**: Importance, Process, Barriers, Tone, Language, Role of Perception and Culture in Communication.<br>• **Project Appraisal**: Market, Technical, and Financial Feasibility.<br>• **Project Financing**: Capital Structure, Sources of Finance, Term Loans, Debentures, Public Issues. | | |
| **Text Books:** | | |
| **T1:** | Robbins, S. P., & Decenzo, D. A., *Fundamentals of Management*, 9th Edition, Pearson Education, 2016. | |
| **T2:** | Koontz, H., O'Donnell, & Weihrich, H., *Essentials of Management*, 9th Edition, Tata McGraw Hill, 2012. | |
| **T3:** | Chandra, P., *Projects: Planning, Analysis, Selection, Implementation & Review*, Tata McGraw Hill Publishing Co, 2014. | |
| **T4:** | Gray, C. F., Larson, E. W., & Joshi, R., *Project Management – The Managerial Process*, 8th Edition, McGraw Hill Education, 2020. | |
| **T5:** | Gido, J., & Clements, J. P., *Successful Project Management*, 6th Edition, Cengage Learning, 2014. | |
| **T6:** | Chandra, P., *Financial Management*, Tata McGraw Hill Publishers, 2014. | |
| **Reference Books:** | | |
| **R1:** | Nicholas, J. M., *Project Management for Business and Technology – Principles and Practice*, Prentice-Hall of India Ltd. | |
| **R2:** | Pinto, J. K., *Project Management – Achieving Competitive Advantage*, 5th Edition, Pearson Publishing Ltd. | |
| **R3:** | Khan, M. Y., & Jain, P. K., *Financial Management*, Tata McGraw Hill Publishers. | |
| **R4:** | Daft, R. L., *Principles of Management*, Cengage Learning, 2009. | |
| **R5:** | Tripathy, P. C., & Reddy, P. N., *Principles of Management*, Tata McGraw Hill, 1999. | |
| **R6:** | Kreitner, R., & Mohapatra, M., *Management*, Biztantra, 2008. | |
| **R7:** | *Management Fundamentals: Concepts, Applications, & Skill Development*, 6th Edition, Sage Publications, 2014. | |
| **Relevant MOOCs Course (Course name and Weblink)** | | |
| 1. Project Management: Planning, Execution, Evaluation and Control, By Prof. Sanjib Chowdhury, IIT Kharagpur<br>Link: https://onlinecourses.nptel.ac.in/noc23_mg124/preview.<br>2. Introduction to Project Management: Principles & Practices, By Dr. Nimisha Singh, Quality Council of India<br>Link: https://onlinecourses.swayam2.ac.in/imb25_mg80/preview. | | |
| **Relevant Topics for Self-study:** | | |
| Arbitration, Conflict Resolution and Project Management Tools | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br>[04051X2]: Multidisciplinary Minor (MDM-2) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **4** | **2** | **L: 2 Hrs./ Week** | **CIE: 20 Marks<br>ISE: 20 Marks<br>ESE: 60 Marks** |
| Refer Annexure-I | | | |
| Refer: Course list | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br>[04051X2]: Multidisciplinary Minor Lab/Tutorial (MDM-2) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **4** | **1** | **P/Tut: 2 Hrs./ Week** | **ESE (P): 25 Marks** |
| Refer Annexure-I | | | |
| Refer: Course list | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br>[04063XX]: Open Elective-II (OE-II) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **4** | **2** | **Tut.: 2 Hrs./ Week** | **ESE: 50 Marks** |
| Refer Annexure-II | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br>[0408203]: Collaborative Skills, Digital Ethics, and Cyber Security (CDC) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **4** | **1** | **P: 2 Hrs./ Week** | **CIE (TW): 25 Marks** |
| Refer: Common Courses | | | |

| Second Year B. Tech (S. Y B. Tech) AY (2025-26)<br>Electronics and Telecommunication Engineering (E&TCE)<br>[0411102]: Indian Constitution and Social Responsibility (ICSR) | | | |
|---|---|---|---|
| **Semester** | **Credits** | **Teaching Scheme** | **Examination Scheme** |
| **4** | **1** | **L: 1 Hrs./ Week** | **CIE (TW): 25 Marks** |
| Refer: Common Courses | | | |

# Annexures

# Annexure-I

## 1. Structure of Multi-Disciplinary Minor Courses

The structure of the Multidisciplinary Minor (MDM) courses is outlined below. However, students must refer to the official MDM course structure and syllabus of the selected domain for the final and detailed framework.

| | | | Teaching Scheme (Hours/Week) | | | | Credits | | | | Examination Scheme and Marks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Course code** | **Name of Course** | **L** | **P** | **T** | **Total** | **L** | **P** | **T** | **Total credits** | **Theory** | | | **Practical** | | | **Semester** |
| | | | | | | | | | | | **CIE** | **ISE** | **ESE** | **CIE** | **ESE** | | **Total** |
| | | | | | | | | | | | **[20]** | **[20]** | **[60]** | **TW** | **P** | **OR** | **550** |
| **3** | **03051X1** | MDM-1 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | **100** |
| **3** | **03052X1** | MDM-1 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| **4** | **04051X2** | MDM-2 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | **100** |
| **4** | **04052X2** | MDM-2 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| **5** | **05051X3** | MDM-3 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | **100** |
| **5** | **05052X3** | MDM-3 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| **6** | **06051X4** | MDM-4 | 2 | - | - | **2** | 2 | - | - | **2** | 20 | 20 | 60 | - | - | - | **100** |
| **6** | **06052X4** | MDM-4 # | - | 2 | - | **2** | - | 1 | - | **1** | - | - | - | 25 | - | - | **25** |
| **8** | **08053X5** | MDM-5 | - | - | 2 | **2** | - | - | 2 | **2** | - | - | - | 50 | - | - | **50** |
| | | **Total** | **8** | **8** | **2** | **18** | **8** | **4** | **2** | **14** | **80** | **80** | **240** | **150** | **0** | **0** | **550** |

**Note:** In course code X is basket number. #: is laboratory or tutorial as per course requirements.

1. Students are required to select one eligible MDM domain at the beginning of Semester III.
2. The total credit requirement for MDM is 14 credits. The maximum marks shall not exceed 125 per semester course, including Theory and CIE/Practical/Oral components.
3. Students must complete all the multidisciplinary minor courses specified under their chosen MDM domain.
4. Change of domain after selection will not be permitted.
5. Students should refer to the last column of the table below to verify their eligibility for selecting a particular MDM domain.

## 2. List of Multi-Disciplinary Minor Domains

| Label | Multi-Disciplinary Minor Domains | SY | | TY | | B-Tech | Offered to students of B Tech Program |
|---|---|---|---|---|---|---|---|
| | | **MD1-1** | **MD2-2** | **MD3-3** | **MD4-4** | **MD5-5** | |
| | | **Sem-III** | **Sem-IV** | **Sem-V** | **Sem-VI** | **Sem-VII/VIII** | |
| **MD1** | **Smart and Sustainable Systems (SSS)** | Fundamentals of Smart and Sustainable Systems (FSSS) & Tut | IoT for Smart and Sustainable Systems (ISSS) & Lab | Data Analytics for Smart and Sustainable Systems Lab (DASSSL) | Security for Smart and Sustainable Systems Lab (SSDL) | Smart and Sustainable System Development & Lab | **ALL** |
| **MD2** | **Finance and Management (F&M)** | Fundamentals of Financial Engineering (FFE) & Tut | Banking, Financial Services and Insurance (BFSI) &Tut | Fundamentals of Stock Market Tutorial (FSMT) | Fintech: Foundations and Applications Tutorial (FFAT) | Financial Derivatives and Risk Management Tutorial (FDRMT) | **ALL** |
| **MD3** | **3D- Printing (3DP)** | 3D modeling and Design (3MD) & Lab | Fundamentals of Additive Manufacturing (FAM)& Lab | Materials and Processes in Additive Manufacturing (MPAM) | Industry 4.0 and Digital Manufacturing (IDM) | Applied 3D Printing and Prototyping Lab (A3DPPL) (Capstone Project) | **ALL** |
| **MD4** | **Electric Vehicles (EV)** | EV foundation – Principles and Concepts (EVPC) & Lab | Advanced Motor Technologies and Power Electronics for EV(AMT) & Lab | EV Powertrain Dynamics and Control System & Lab (PDCL) | Intelligent EV Systems:AI Iot and Automation Lab (IEVL) | Capstone Project in Electric Mobility Lab (CPEML) | **ALL** |
| **MD5** | **Applied Mathematics for Engineering (AME)** | Linear Algebra with Python & Lab | Statistical Techniques and Numerical Methods with R & Lab | Graph Neural Networks with Python & Lab (GNNPL) | Mathematical Optimization Techniques using R & Lab (MOTRL) | Field Study/Case Study Lab (FSL/CSL) | **ALL** |
| **MD6** | **Software Development (SD)** | Data Structures and Algorithms (DSA) & Lab | Object Oriented Programming (OOP) &Lab | Database and Management Systems & Lab (DBMSL) | Web Development & Lab (WDL) | System Programming and Operating Systems Lab (SPOSL) | **Only E&TCE** |
| **MD7** | **Autonomous and Intelligent Systems (AIS)** | Digital Systems and Organization (DSO) & Lab | Smart System Engineering (SSE) & Lab | Embedded IoT Systems & Lab (EISL) | Autonomous Systems & Lab (ASL) | Capstone Project# | **All except E&TCE** |
| **MD8** | **Embedded Systems (ES)** | Fundamental of Microcontroller (FM) & Lab | Embedded Processors –I (EP -I) & Lab | Microcontrollers and IoT & Lab (MIL) | Embedded System Design and RTOS & Lab (ESD-RTOSL) | Capstone Project using Microcontrollers lab (CPML) | **All Except E&TCE** |
| **MD9** | **AI & Machine Learning (AI-ML)** | Statistical Data Analysis & Lab | Machine Learning (ML) & Lab | Artificial Intelligence & Lab (AIL) | Next Gen AI & Lab (NGAIL) | Deep Learning (DL) | **Only E&CE** |

Link: **Detailed Syllabus**

# Annexure -II

## 1. Guidelines for Open Elective Courses

i. Open Elective – I will be offered in third semester as foreign language as prescribed in the structure.

ii. Open Electives – II, III, IV will be offered through SWAYAM/ NPTEL MOOCs of Equivalent Credits.

iii. Departments shall prepare the baskets of open elective courses from discipline/faculty other than respective major programs. Students may choose any course from the basket without adhering to any one stream.

iv. Credits & Grade will be awarded based on the Marks Obtained through the certification including assignments and proctored examination as per the MOOCs Policy.

| | | | Teaching Scheme (Hours/Week) | | | | Credits | | | | Examination/ Evaluation Scheme and Marks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sem | Course code | Name of the Course | L | P | T | Total | L | P | T | Total | Theory | | | Practical | | | Total |
| | | | | | | | | | | | CIE [20] | ISE [20] | ESE [60] | CIE TW | ESE P | ESE OR | |
| **3** | **OE-I** | **Foreign Language Studies (FLS)** | **-** | **-** | **2** | **2** | **-** | **-** | **2** | **2** | **-** | **-** | **-** | **50** | **-** | **-** | **50** |
| **4** | **OE-II** | **MOOCs** | **-** | **-** | **2** | **2** | **-** | **-** | **2** | **2** | | | **50** | **-** | **-** | **-** | **50** |
| **5** | **OE-III** | **MOOCs** | **-** | **-** | **2** | **2** | **-** | **-** | **2** | **2** | **-** | **-** | **50** | **-** | **-** | **-** | **50** |
| **6** | **OE-IV** | **MOOCs** | **-** | **-** | **2** | **2** | **-** | **-** | **2** | **2** | **-** | **-** | **50** | **-** | **-** | **-** | **50** |

## 2. Guidelines for MOOCs

1. The department shall release a list of approved SWAYAM-NPTEL courses before the commencement of every semester.
2. Students shall register for the approved Courses as per the schedule announced by SWAYAM-NPTEL.
3. A student shall undergo the courses only from the list notified by the department through SWAYAM/NPTEL platform and complete all the assignments and examination requirements as specified by SWAYAM/NPTEL.
4. SWAYAM-NPTEL Courses are considered for transfer of credits only if the student concerned has successfully completed and obtained the SWAYAM-NPTEL Certificate.
5. The credit equivalence for SWAYAM-NPTEL Courses: 12 weeks – 3credits; 8 weeks – 2 credits; 4 weeks – 1 credit.
6. Equivalent marks will be considered for awarding the grades as specified in examination rules and regulations. The weightage for assignments is 40%, while the weightage for the proctored examination will be 60% for award calculating SGPA/CGPA. Students must score a minimum of 40% of the total marks by combining both assignments and proctored examinations

7. A student must submit the original SWAYAM-NPTEL Course Certificates to the Head of the Department concerned, with a written request for the transfer of the equivalent credits. On verification of the SWAYAM-NPTEL Course Certificates and approval by the head of the department, credits will be awarded.
8. The Institute shall not reimburse any fees/expenses a student may incur for the SWAYAM-NPTEL Courses.
9. If the SWAYAM/NPTEL course calendar does not align with the institute's calendar, the department shall facilitate and conduct examination of the relevant course of equivalent credits in physical/virtual mode and award the credits accordingly.

## Annexure -III

**Relevant Topics for Self-study**

**These topics are optional and suggested for students for self-study to get additional exposure. However, these topics will not be considered for any evaluation and credits.**